

Fitbit Activity Analysis

Understanding how activity metrics reflect effort, calories, and behavior

by Vladyslav Yakymchuk

Data Source & Context

Dataset: Fitbit Fitness Tracker Data

Source: Kaggle (public dataset)

License: CC0 (Public Domain – free to use by anyone)

Dataset Overview

This dataset contains anonymized data collected from Fitbit users, including daily activity levels, sleep patterns, and basic health-related metrics. The data reflects real-world usage of consumer fitness trackers.

Data Scope

- Time period: April – May 2016
- Participants: 30 Fitbit users
- Granularity: Daily-level records

Data Limitations

- Small sample size
- Data collected from consumer-grade devices
- Users are not representative of the general population

Project Goals



MET (Metabolic Equivalent of Task) measures energy use relative to rest



Understand what drives calorie burn



Explore daily and weekly activity patterns

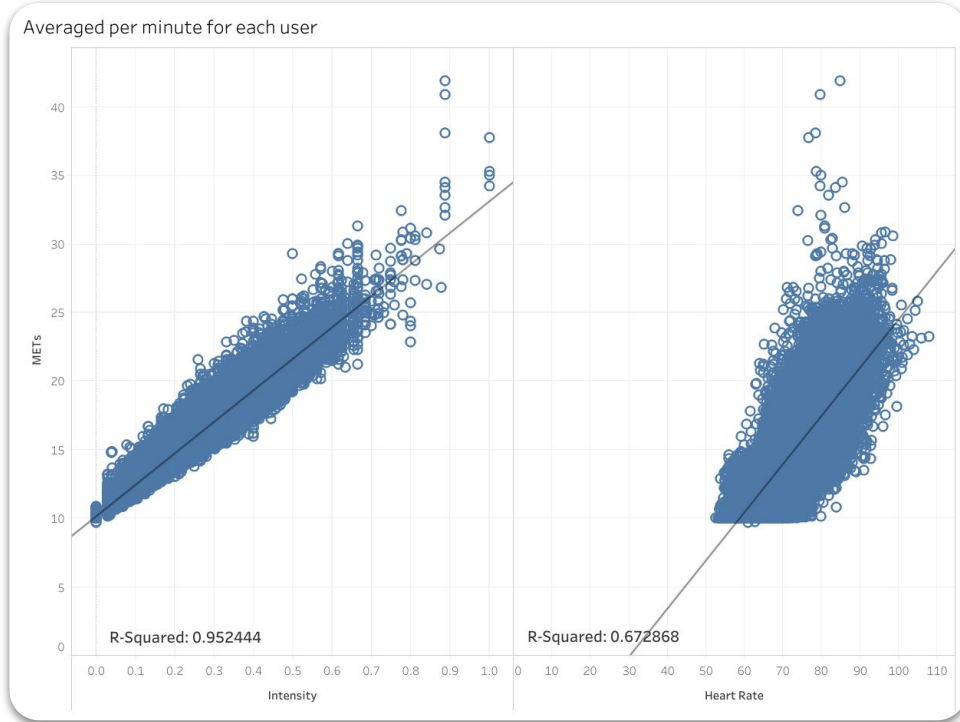


Examine the link between activity and sleep

What is MET?

- MET (Metabolic Equivalent of Task) measures energy use relative to rest
- 1 MET = resting energy expenditure
- Higher MET = higher physical effort
- Used to standardize activity intensity

What determines METs?



Key result:

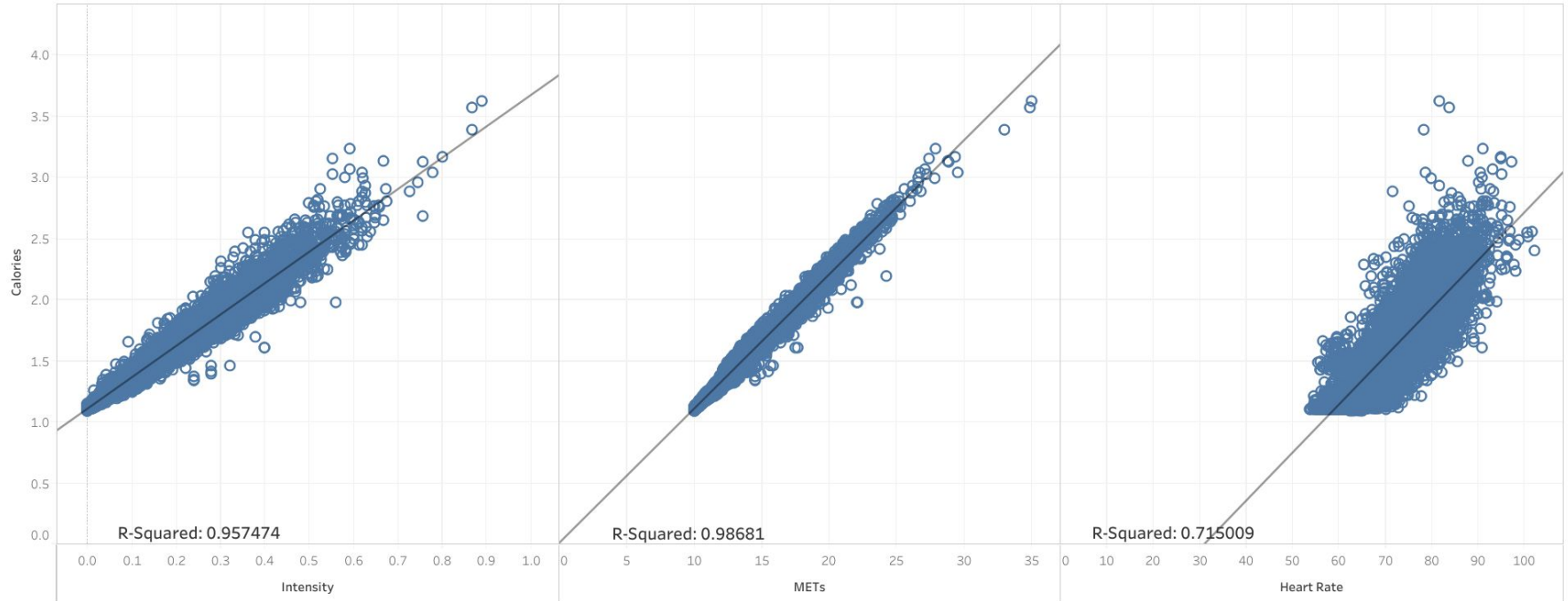
- MET vs Intensity: very strong relationship ($R^2 \approx 0.95$)
- MET vs Heart Rate: weaker relationship ($R^2 \approx 0.67$)

Interpretation:

- Intensity almost fully determines MET
- Heart rate reflects effort less directly

What Predicts Calories Burned?

Averaged per minute for each user



What Predicts Calories Burned?

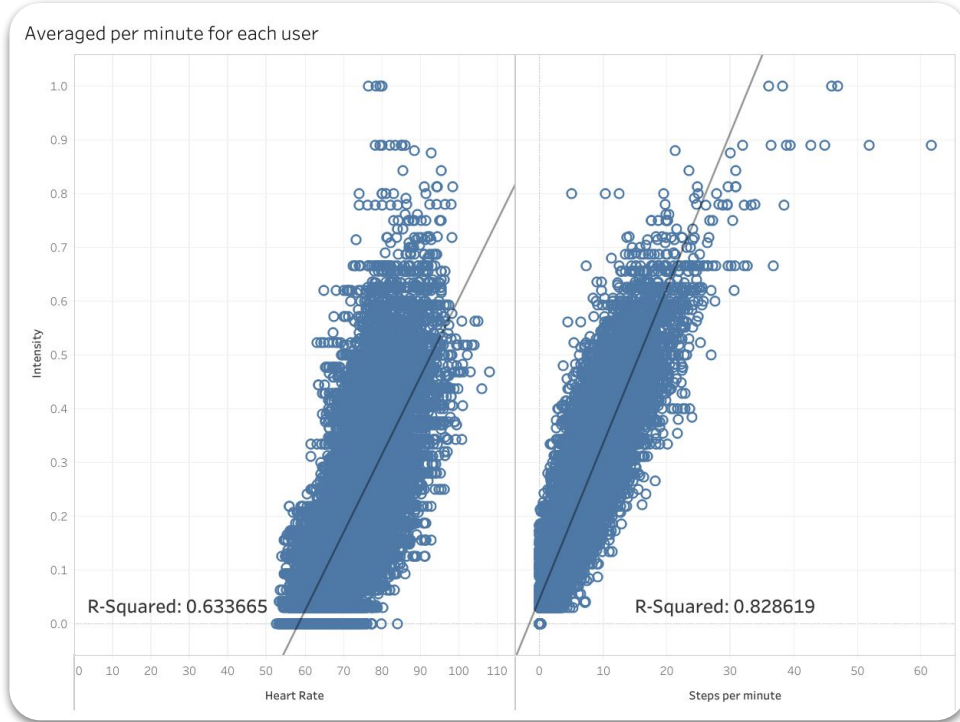
Results:

- METs vs Calories: near-perfect relationship ($R^2 \approx 0.99$)
- Intensity vs Calories: very strong ($R^2 \approx 0.96$)
- Heart Rate vs Calories: noticeably weaker ($R^2 \approx 0.72$)

Conclusion:

- Calories are fundamentally driven by METs and Intensity
- Heart rate adds context but is less stable

What determines Intensity?



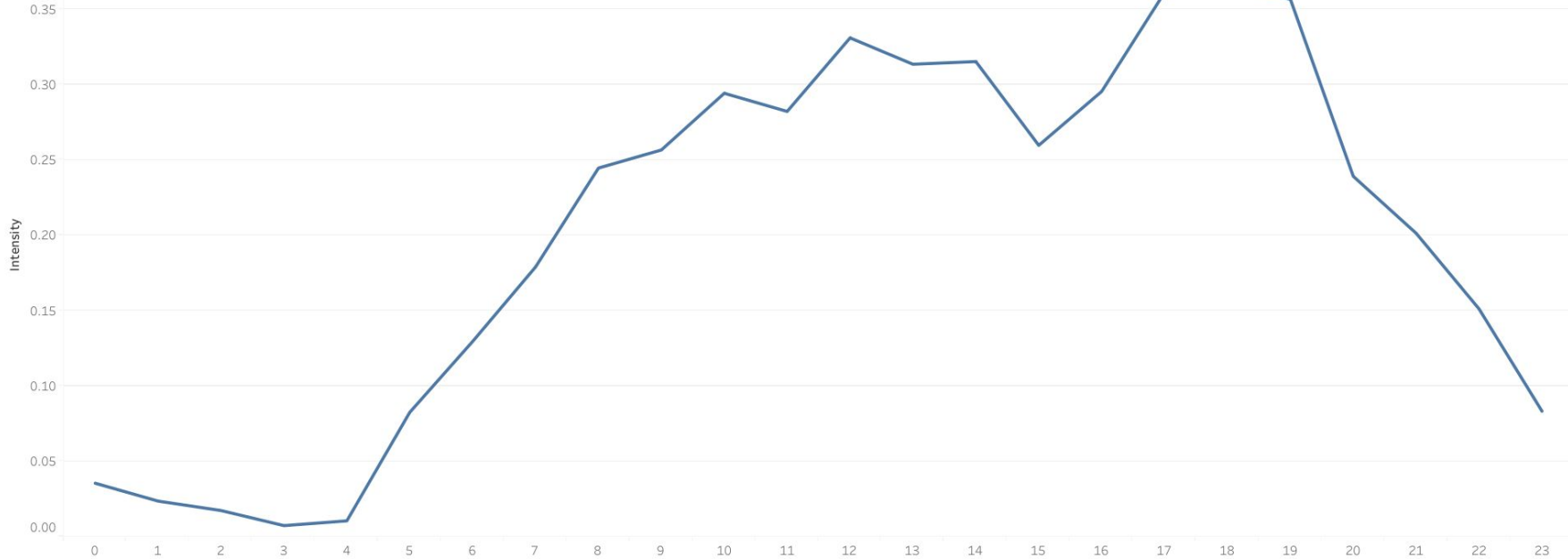
- Intensity vs Heart Rate: moderate relationship ($R^2 \approx 0.63$)
- Intensity vs Steps per Minute: strong relationship ($R^2 \approx 0.83$)

Insight:

- Intensity depends more on movement than physiology

Activity by Time of Day

Averaged per minute for each user



Activity by Time of Day

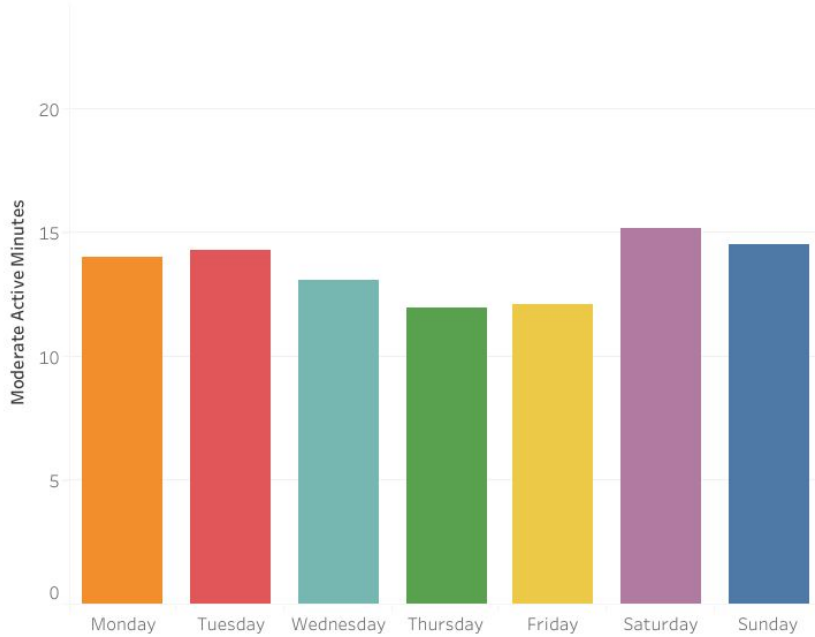
- Lowest activity at night (sleep period)
- Gradual rise in the morning
- Peak activity around 5–6 p.m.
- Decline in the evening

Interpretation:

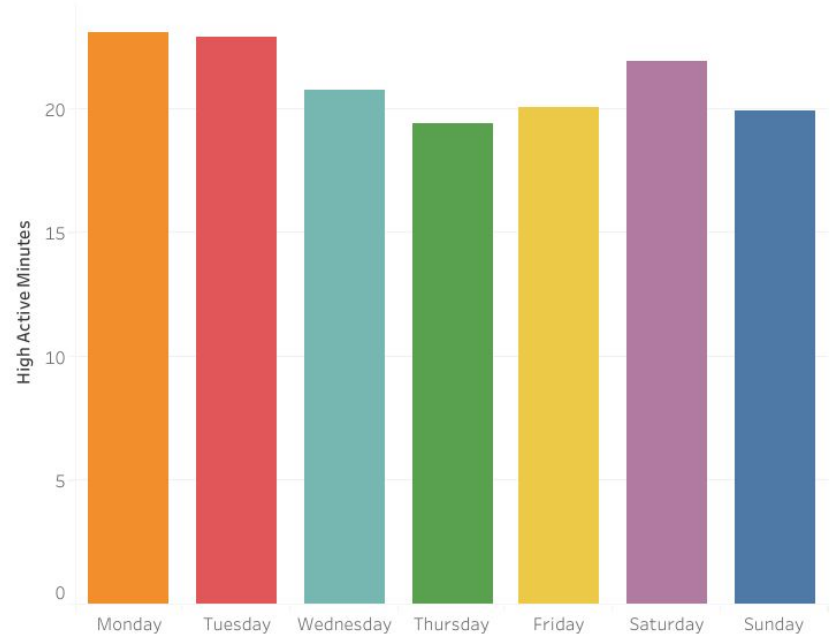
- Activity follows daily routines rather than random behavior

Activity by Day of Week

Averaged per minute for each user



Averaged per minute for each user



Activity by Day of Week

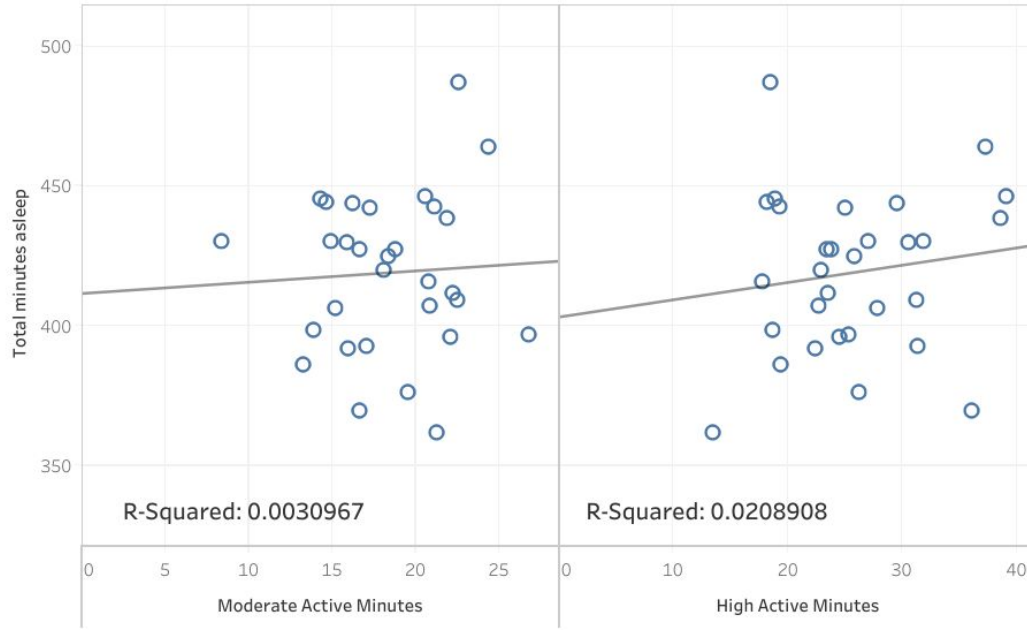
- Activity levels are relatively stable
- Slight increase on weekends (especially Saturday)
- Lowest activity on Thursday and Friday

Insight:

- Users maintain consistent baseline activity

Does Activity Affect Sleep?

Averaged per minute for each user

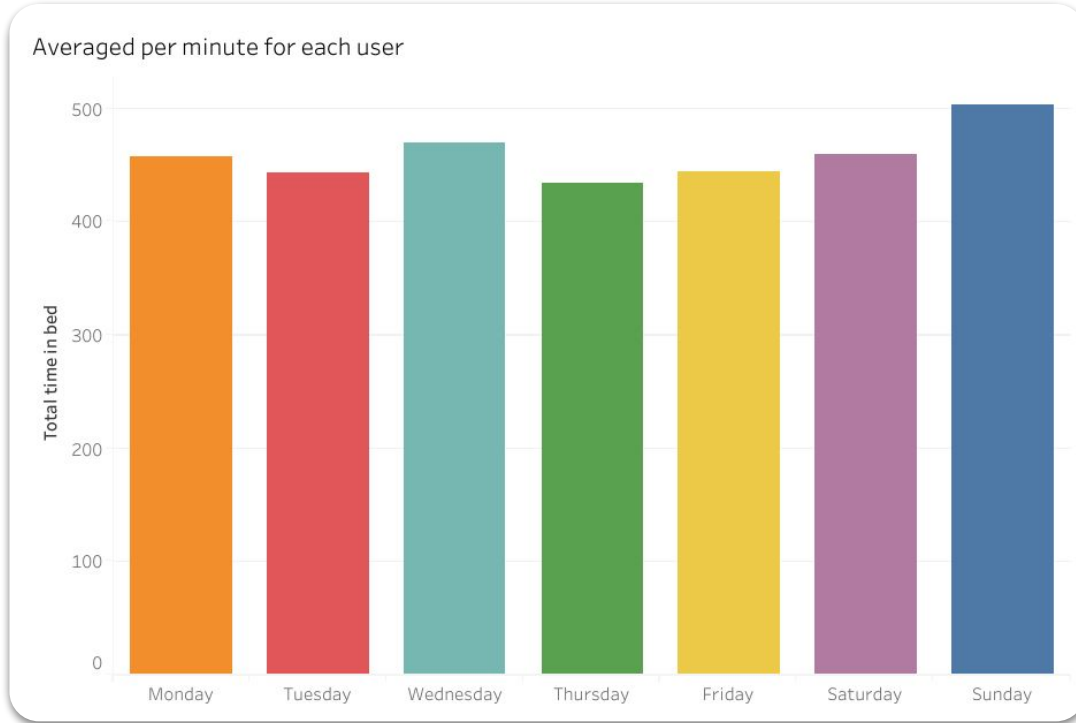


- Sleep vs Moderate Activity: no correlation ($R^2 \approx 0.00$)
- Sleep vs High Activity: very weak correlation ($R^2 \approx 0.02$)

Conclusion:

- Sleep duration is largely independent of prior-day activity

Time in Bed by Day



- Longest time in bed on Sundays
- Slight mid-week increase (Wednesday)

Interpretation:

- Likely reflects recovery and accumulated fatigue

Key Takeaways

- METs and Intensity are core drivers of energy expenditure
- Heart Rate is informative but less reliable
- Activity follows strong daily and weekly patterns
- Sleep duration shows no strong linear link to activity

Overall conclusion:

Fitbit data models physical activity effectively, prioritizing observable movement over individual physiological variability.